



Humanoid robots face key challenges

From battery life to demand, the industry's not growing as fast

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Over the past few years, emerging firms in robotics have made bold bets that humanoid robots will reshape manufacturing, logistics, and even everyday life. Backed by eye-popping valuations and massive investment rounds, companies such as Agility Robotics, Tesla, and Figure are laying out plans to deliver tens of thousands, even hundreds of thousands, of

humanoid robots. But recent reporting shows that the road from demos to mass deployment runs straight into some severe practical hurdles.

Here are the core challenges the industry must clear and why battery life, reliability, safety, and demand sit at the heart of the problem.

Scaling expectations vs reality
Manufacturers are making serious

projections, Agility Robotics hopes to ship “hundreds” of its Digit robots in 2025 with a factory capable of over 10,000 units a year. Tesla is targeting 5,000 Optimus units in 2025, escalating to 50,000 in 2026. Analysts are just as ambitious; Bank of America forecasts ~18,000 humanoid units shipped in 2025; Morgan Stanley sees more than 1 billion units by 2050, in a market worth some US\$5 trillion.

But those numbers are almost entirely hypothetical. So far, most deployments are small trial projects under controlled conditions. Multi-purpose, general-purpose humanoids that can safely and reliably operate in messy, unpredictable settings are still under development.

Battery life: It's more than just a number

Among the various engineering challenges, battery life is one of the most concrete and the easiest to quantify, but also one of the hardest to engineer for real-world use. A robot that must spend much of its time charging is no better than a very expensive prop, if you think about it.